YouthTeamUp

CERTIFICATE

OF PARTICIPATION

GIVEN TO

Fatma Abdallah

For participating in the 4th YouthTeamUp ***Deep Virtual Exchange (DVE)*** *program which was held from 15th March 2026 to 8th June 2026, and going through technical training in the following areas:*

- Hardware and communication:
  - Introduction to Internet of Things, Electronics for Internet of Things
  - Development boards and peripherals
  - Wireless communication essentials
  - LoRa radio communication
- Waziup technologies:
  - LoRaWAN End Devices (WaziDev, WaziSense)
  - WaziGate, WaziCloud, Waziup APIs
- Web and mobile application development
  - IoT Web Apps, HTTP REST APIs
  - MQQT APIs, Databases, cloud platforms
- Data analysis, visualization and Artificial Intelligence (AI)
  - Fundamentals of AI, Machine Learning
  - Computer Vision, Data visualizations
  - IoT data analysis
- Business model design.



In collaboration with other team members ***Fatma Abdallah*** *has also successfully developed a Global Impact Project (GIP) with the title:*

„ DIVERSE GROW – Smart Agriculture Cooperative Platform “

Dr. Omid Fatahi Valilai
Project Coordinator YouthTeamUp

11th June 2026

CONSTRUCTOR UNIVERSITY

BongoHive DTBi edulife Fondazione ETS WAZIUP

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